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LONG PAPER

Design for educational equity: a collaborative remote Braille learning system and tactile intervention for visually impaired children

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Abstract
Blind children in remote mountainous areas suffer from a lack of specialized teachers and learning tools. Traditional distance education often fails to address their specific need for tactile interaction and real-time feedback. This study aims to design and empirically evaluate a collaborative remote Braille learning system to promote educational equity for visually impaired children in underserved areas. This study integrates Quality Function Deployment (QFD) and the Function-Behavior-Structure (FBS) model to bridge the gap between user needs and engineering design. Through interviews with 20 blind students, 16 parents, and 4 teachers, user needs were weighted and translated into functional specifications. Validation via the Critical Incident Technique (CIT) demonstrated that the system's optimized structural features and real-time audio-tactile feedback loops significantly reduced the cognitive load associated with tactile decoding. This feedback mechanism enabled independent error correction, a critical capability previously unattainable in traditional remote settings. Furthermore, the supporting mobile applications successfully bridged the literacy gap for sighted parents, facilitating data-driven tutoring and continuous monitoring in home environments. This study presents a scalable, low-cost solution to advance remote Braille literacy. While QFD and the FBS model served as effective frameworks for translating complex user needs into engineering parameters, the primary contribution lies in the realization of a synchronized haptic-digital ecosystem. This approach significantly mitigates the geographic and sensory barriers faced by visually impaired children in under-resourced regions, promoting broader educational equity.

Keywords Inclusive design · Visually impaired children · Remote braille education · QFD & FBS model · Critical incident technique · Educational equity

1 Introduction

Blind children are highly dependent on tactile exploration of the environment due to their visual impairment, but their tactile cognitive abilities differ from those of ordinary children [1], and are often accompanied by other sensory impairments, which further constrain learning development [2]. Remote mountainous areas are limited by geographic and economic factors, with a lack of Braille education resources and insufficient policy support, resulting in a low education penetration rate [3]. However, most of the existing programs rely on high-cost technologies or network conditions, making it difficult to adapt to remote areas.

Limited access to Braille literacy not only hinders academic progress but also imposes severe emotional and cognitive burden on visually impaired children. Recent systematic reviews indicate that children with visual impairments (VI) who lack adequate literacy support are significantly more likely to experience academic underachievement compared to their sighted peers, primarily due to slower reading speeds and reduced comprehension without tactile reinforcement [2]. Beyond academics, studies highlight a profound correlation between illiteracy in the blind community and adverse mental health outcomes. For instance, a 2025 analysis of national health data revealed that children with visual

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impairments exhibit significantly higher rates of anxiety (64%) and depression (44%) compared to those without impairments, with cognitive processing difficulties being a contributing factor [4].

Furthermore, the inability to engage with educational materials independently often leads to reduced self-efficacy and heightened social isolation. Szekely et al. [5] argue that without accessible assistive technologies to bridge the literacy gap, visually impaired youth face increased "technological overwhelm" and social stigma, further diminishing their confidence in social interactions. In rural contexts, this manifests as a "double disadvantage", where geographic remoteness combines with sensory impairment to create severe barriers. Alsamiri et al. [6] emphasize that in such resource-constrained environments, the lack of specialized educational support (e.g., accessible textbooks and trained teachers) leads to a complete detachment from social life. This exclusion creates a vicious cycle of poverty and marginalization, as low literacy levels in childhood directly predict limited employment opportunities and economic dependency in adulthood, particularly in underserved rural regions [7].

The primary objective of this study is to design and evaluate a low-cost, multi-terminal remote Braille learning ecosystem tailored for visually impaired children in underserved rural areas. While existing assistive technologies have made strides in accessibility, significant gaps remain in addressing the need for real-time haptic feedback and professional supervision in remote home-learning contexts. To bridge the gap between abstract user needs and concrete engineering solutions, we utilized Quality Function Deployment (QFD) and the Function-Behavior-Structure (FBS) model as our methodological framework. Moving beyond this methodological integration, the core contributions of this paper are three-fold: (1) the optimization of tactile parameters tailored to the physical and cognitive needs of early-age Braille learners; (2) the development of a collaborative software-hardware ecosystem that empowers both parents and teachers; and (3) a qualitative validation demonstrating how remote haptic interventions foster independent learning and educational equity in resource-constrained environments.

2 Literature

2.1 Global approaches to inclusive education

Strategies to support the education of visually impaired children vary significantly across economic contexts, creating a distinct dichotomy between developed and developing regions [8]. In the Global North, inclusive education often relies on advanced assistive technologies and integrated policy frameworks. Recent policy analyses by the [9] emphasize that developed education systems are increasingly mandating "twin-track" approaches that combine mainstream schooling with specialized digital accessibility standards, ensuring that learners with disabilities are not merely "placed" in schools but are supported by systemic infrastructure. Furthermore, the integration of sophisticated tools, such as AI-driven screen readers and refreshable Braille displays, has become a standard pedagogical expectation in these regions, fundamentally reshaping the learning landscape for visually impaired students [10].

Conversely, in the Global South, support systems are frequently driven by NGOs and specific government interventions aimed at basic resource provision, often struggling to move beyond the foundational level of survival to educational empowerment. Le Fanu et al. [11] notes that, in regions like Sub-Saharan Africa, the primary challenge remains the severe shortage of specialized teachers and accessible learning materials, which forces inclusion efforts to focus on basic literacy rather than digital proficiency. This "digital divide" is further exacerbated by a lack of institutional training; Talafhah and Bataineh [12] highlights that even when assistive technologies are donated or introduced, they often fail to improve learning outcomes due to insufficient teacher training and compatibility issues with local curricula. However, recent studies suggest that without addressing the deeper "barriers to implementation", such as the lack of continuous technical support and the high cost of device maintenance, these sporadic interventions may not sustain long-term educational equity [13].

While these initiatives tackle the fundamental lack of resources, they often struggle to overcome the "geographic barrier" in remote areas. Distance education has emerged as a critical solution to provide opportunities for groups outside the traditional system, utilizing technology to overcome multiple challenges [12]. Research indicates that distance education can be as effective as traditional methods when properly implemented, with each having its own strengths in contributing to the learning experience [13]. However, a significant gap remains: most successful distance learning models rely on high-bandwidth infrastructure or expensive hardware, rendering them inaccessible to the specific demographic of this study—blind children in remote, economically underdeveloped villages [14, 15]. Furthermore, the high cost of commercial assistive devices often exacerbates this exclusion, creating a "secondary digital divide" for low-income users [10]. The challenge, therefore, is not

merely providing "education", but designing low-cost, high-access delivery systems that can function effectively within these resource-constrained environments [16].

2.2 Critique of current remote learning tools

The evolution of distance learning tools for the visually impaired has progressed from print and broadcast to computer-mediated communication [17, 18]. Following the COVID-19 pandemic, scholars have introduced various innovative pedagogies, yet each presents specific limitations when applied to remote Braille literacy [19].

Digital and Virtual Solutions: Recent innovations have focused heavily on digital simulation [20]. For example, virtual haptic perception tools have been proposed to teach 3D geometry and help children understand dynamic objects through force-feedback devices [21]. Mobile applications incorporating Text-to-Speech (TTS) and voice recognition have also been optimized to provide multi-sensory approaches for mathematical skills and general STEM accessibility [22]. While these digital tools enhance auditory and cognitive engagement, they suffer from a "tactile deficit" in the context of literacy. Research emphasizes that audio-based substitution cannot physically simulate the dot-matrix structure of Braille, which is essential for developing the fine motor skills required for reading and writing [10].

Physical and Tangible Tools: To address tactile needs, physical aids such as tangible programming blocks have been developed to support inclusive education [23]. Innovations like LEGO "Braille Bricks" and similar tangible coding kits (e.g., BATS) have successfully lowered the entry barrier for sighted peers to participate, fostering social inclusion [24]. However, these low-tech tools often lack connectivity, making them unsuitable for remote instruction where teacher feedback is absent. Conversely, electronic tools like the "BecDot" or newer low-cost refreshable devices offer gamified approaches integrating light and sound [16]. Yet, a critical flaw in many such devices is their inability to support dynamic content updates or their reliance on high-cost peripherals, limiting their long-term educational value.

The Design Gap: A critical analysis of existing literature reveals a functional gap: high-tech solutions offer dynamic content but are prohibitively expensive for rural families, while low-tech solutions are affordable but lack the remote connectivity and real-time feedback mechanisms necessary for distance learning [24]. This "secondary digital divide" significantly hampers access for users in underserved regions [12]. The present study aims to fill this specific void by developing a system that is both cost-effective for remote villages and technologically capable of supporting dynamic, remote Braille instruction [15].

2.3 Theoretical framework: bridging inclusive needs and engineering structure

While distance education has democratized access to learning, it often fails to accommodate visually impaired (VI) learners, leading to a "digital exclusion" where standard platforms remain incompatible with screen readers and haptic needs. Recent studies indicate that remote learning during the pandemic exacerbated these disparities, as VI children struggled to engage with visual-centric curricula without physical guidance. Current inclusive design theories emphasize that the needs of VI children are often "tacit" and deeply embodied, making them difficult to capture through traditional interviews alone [25]. In the context of remote villages, this challenge is compounded by a chronic shortage of specialized teachers who could otherwise interpret these needs or adapt materials locally [26]. Consequently, a standard "trial-and-error" design process is insufficient; a structured, human-centered methodology is required to accurately translate these ambiguous, context-dependent needs into precise engineering specifications [27]. Such rigorous frameworks are essential for developing low-cost, multifunctional devices that are robust enough for resource-constrained environments.

2.4 Rationale for integrating QFD and FBS

To bridge the gap between abstract educational needs and concrete hardware solutions, this study integrates Quality Function Deployment (QFD) with the Function-Behavior-Structure (FBS) model.

Why QFD? (The Translation Tool): In inclusive design, the "Voice of the Customer" (VoC) often contains emotional and sensory requirements that engineers find hard to quantify (e.g., safe feeling or fun interaction). QFD provides a systematic matrix to map these qualitative needs (e.g., Parent's need for monitoring or Student's need for safety) directly into quantitative technical characteristics. This ensures that the specific constraints of the remote mountain environment, such as cost and durability,are not lost during the design process.

Why FBS? (The Materialization Tool): While QFD defines what to do, it does not explain how to construct it physically. The FBS model is particularly effective for designing novel assistive devices because it forces a separation between Function (the goal), Behavior (the interaction), and Structure (the physical form). This is critical for VI devices where the Structure (e.g., the height of a Braille dot) directly dictates the Behavior (readability).

By coupling these two models, we create a closed-loop design rationale: QFD ensures the design aims for the right

educational equity goals, while FBS ensures the hardware achieves the necessary haptic affordances [28].

3 Methods

3.1 Overview of the research framework

This study adopts a systematic, evidence-based design framework to develop a remote Braille learning ecosystem for visually impaired children in under-resourced regions [29]. The research process is operationalized through five distinct yet interconnected phases: user needs identification, technical requirement translation, conceptual system design, prototype development, and empirical evaluation.

As illustrated in Fig. 1, the framework initiates with field research to capture raw multi-stakeholder requirements. Subsequently, Quality Function Deployment (QFD) is employed to transform these qualitative needs into quantifiable technical targets. The Function-Behavior-Structure (FBS) model is then applied to map functional logic onto specific hardware and software architectures. The process culminates in the development of a functional prototype, which is validated through the Critical Incident Technique (CIT) to ensure the design effectively addresses real-world educational barriers.

3.1.1 Participants

The study employed a purposive sampling strategy. A total of 20 visually impaired studentswere recruited based on the following inclusion criteria: (1) medically diagnosed with complete blindness or severe low vision; (2) aged between 6 and 15 years (compulsory education stage); (3) residing in rural or semi-rural areas with limited access to special education resources. Additionally, 16 parents (primary caregivers) and 4 certified Teachers of Students with Visual Impairments (TVIs) working at the school for the blind were invited to provide triangulation data.

Interviews: The author invited 4 blind school teachers, 16 parents of blind children, and 20 visually impaired students (aged 10–15) from XXX School for the Blind to participate in the questionnaire research on the weights of user needs. The inclusion criteria were indeed children aged 6–15; however, the 20 students who ultimately participated in the final evaluation were aged 10–15. The interviews with adult blind students reflected the user needs of some of the blind children themselves for the preschool teaching system by recalling their own teaching experiences and teaching needs during their preschool years. Teachers were invited to participate in the user interviews with the aim of understanding the user needs of the preschool teaching system from the perspective of teachers.

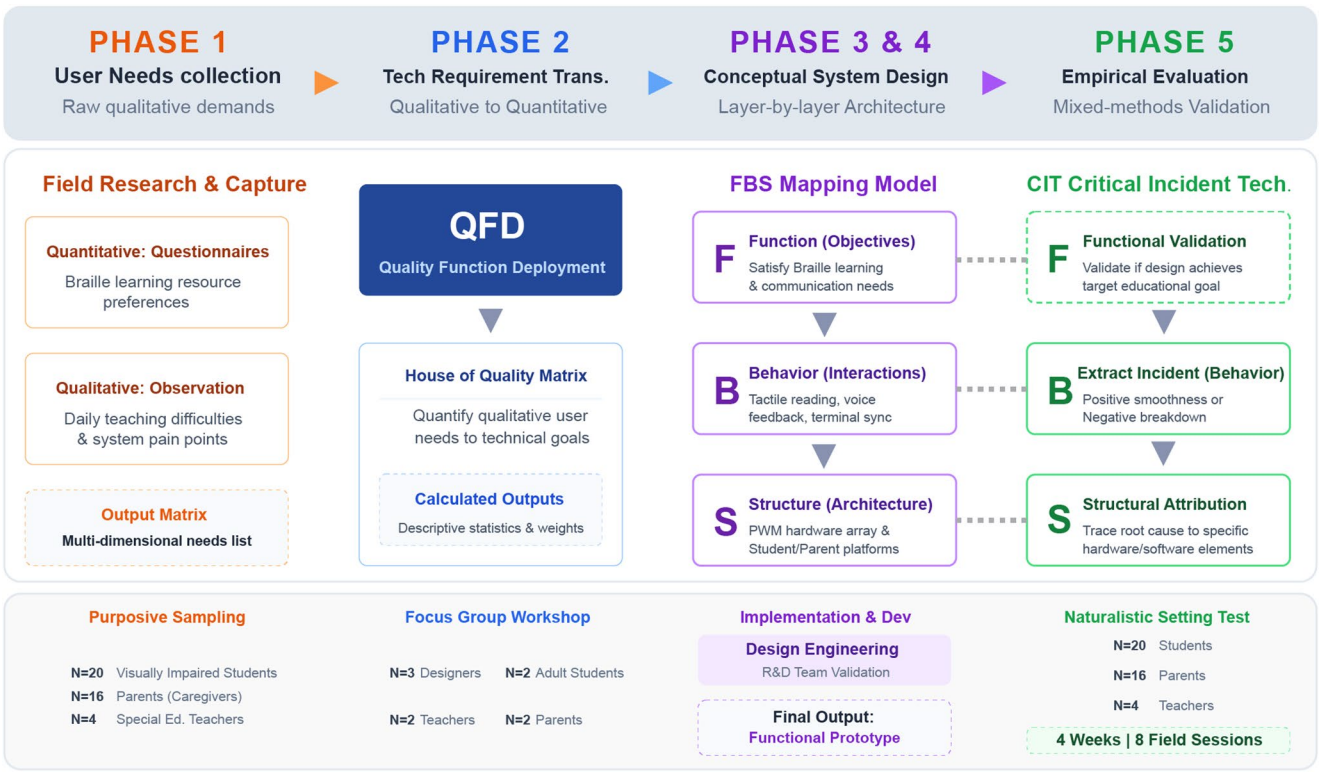


Fig. 1 Research progress diagram

Focus group: The author invited 3 professional industrial designers, 2 blind school teachers, 2 parents of blind children, and 2 adult blind students to hold focus group discussions to transform user needs into corresponding technical features.

CIT participants: The evaluation involved the same stakeholder groups defined in the needs analysis phase to ensure consistency 20 visually impaired students (aged 10–15) from XXX School for the Blind, representing the primary end-users. For qualitative usability and Critical Incident Technique evaluations, research indicates that a sample size of 15–20 participants is highly effective for reaching data saturation and identifying over 90% of critical usability issues in human–computer interaction contexts´[30]. In addition, 16 parents and 4 special education teachers participated to evaluate the supporting software ecosystem (Parent/Teacher Terminals). The testing was conducted in a naturalistic setting (classroom and simulated home environment) over a period of 4 weeks, with a total of 10 h of in-depth observation and 8 one-hour field sessions.

The study was ethically reviewed by the academic committee of XXX University. Informed consent was obtained from all subjects who participated in this study. We confirmed that all experiments were conducted according to relevant guidelines and regulations. Before data collection, we verbally informed participants that the assessment was anonymous, stated that the assessment was for research purposes only, informed them of the main content of the study, and obtained their consent.

3.1.2 Overview of data analysis

To ensure a rigorous evaluation, this study employed a mixed-methods approach to data analysis. Quantitative data derived from the user needs surveys (the QFD phase) were analyzed using descriptive statistics, specifically means and rankings, to prioritize technical characteristics. For the qualitative data gathered during the CIT observations and interviews, we applied a deductive coding strategy mapped directly onto the FBS framework. By tracing observed user behaviors (Behavior) back to specific physical or digital design elements (Structure), we validated whether the intended educational objectives (Function) were successfully achieved. This analytical process ensures a robust and traceable link between initial user needs and the final system evaluation.

3.2 System design and development (phases 1–4)

The system was developed through a rigorous four-phase design process (Phases 1–4) translating raw user needs into engineering specifications [31]. The data obtained from the field research were categorized and organized, and a questionnaire on the user needs of Braille preschool education product design was designed (Appendix A). We employed Quality Function Deployment (QFD) to quantify stakeholder requirements (e.g., safety, cost, interactivity) into technical targets [32]. Subsequently, the Function-Behavior-Structure (FBS) model was utilized to map these technical goals onto specific hardware and software architectures [33–36] (Note: To maintain the manuscript's focus on the empirical evaluation, the comprehensive details of the system design and development process covering Phases 1 through 4—including the QFD weighting matrices, FBS functional mapping, prototype iterations, and tactile parameter optimization experiments—are provided in Appendix D).

3.3 Phases 5: empirical evaluation

3.3.1 Theoretical rationale for CIT

Methodological Suitability for Special Demographics Evaluating interactive systems for visually impaired children presents unique challenges that standard quantitative metrics (e.g., Likert scales) often fail to address. These traditional methods frequently overlook the nuanced, tacit needs of this demographic, such as tactile comfort and psychological safety, by relying on aggregated data. To bridge this gap, this study adopts the Critical Incident Technique (CIT) as the dominant qualitative approach. Unlike questionnaires that provide average satisfaction scores, CIT excels in capturing specific, contextualized "moments of truth"—distinct behavioral breakthroughs or obstacles—that fundamentally drive user behavior and learning outcomes. This methodological shift allows researchers to move beyond statistical averages to focus on the discrete interactions that define the user experience.

A primary advantage of CIT lies in its ability to provide deep insights into complex cognitive and emotional processes that surveys often miss. In educational contexts, Joshi [37] demonstrates that CIT effectively reveals nuanced problem-solving behaviors. Similarly, Nazari and De Costa [38] highlight its unique ability to uncover specific turning points in identity development, while Liu [39] validates its use in detailing coping strategies that standardized scales fail to articulate. Furthermore, CIT demonstrates superiority in product usability and safety validation by identifying critical interaction points. Shekhar et al. [40] illustrate that determining the root causes of specific "critical incidents" is essential for rigorous safety validation. Because quality is defined by specific encounters rather than average performance, isolating these incidents allows designers to target optimization precisely, significantly enhancing user satisfaction [41].

3.3.2 Data collection procedure

Data was triangulated from three sources to ensure validity:

1. Direct Observation: Researchers documented real-time interactions using "unstructured field notes," focusing on the students' hand gestures, facial expressions, and latency in completing tasks.
2. Task-Based Assessment: Participants were asked to perform core tasks defined by the QFD technical characteristics, such as "Braille input", "Live broadcast connection", and "Homework review".
3. Debriefing Interviews: Short, semi-structured interviews were conducted immediately after tasks to capture subjective feedback from teachers and parents regarding the system's usability.

3.3.3 Data analysis: the FBS mapping protocol

To move beyond descriptive storytelling, the collected observation logs were analyzed using a deductive coding method based on the FBS model. The analysis followed a three-step protocol:

Step 1: Incident Extraction. Critical Incidents were identified from the logs—defined as moments where the user either achieved a goal with exceptional smoothness (Positive Incident) or encountered a breakdown requiring assistance (Negative Incident).

Step 2: Structural Attribution. Each incident was traced back to a specific physical or digital design element (Structure), such as the "1.5 mm dot height" or the "Voice Feedback Loop".

Step 3: Functional Validation. We verified whether the observed Behavior (B) confirmed the realization of the intended Function (F), thereby validating the design hypothesis established in the theoretical framework.

4 System overview

Before detailing the evaluation, a brief overview of the developed system is necessary. The architecture (detailed in Fig. 2) adopts a four-layer model comprising a tactile hardware terminal for visually impaired students, and supporting mobile applications for parents and teachers. The system enables real-time remote haptic intervention and data-driven tutoring. Detailed hardware schematics and software interfaces can be found in Appendix D.

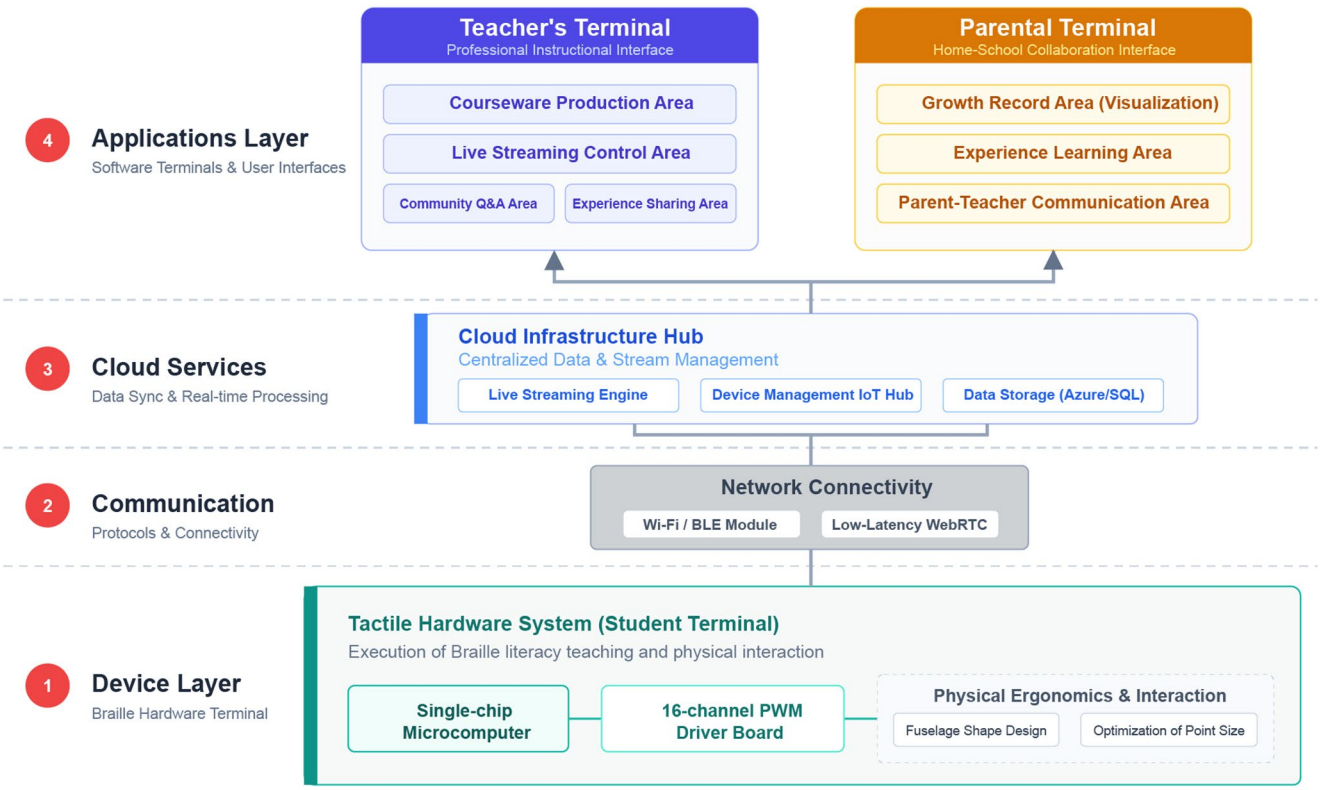


Fig. 2 Block diagram of the learning machine system architecture

5 Results

To validate the system's effectiveness in multi-party collaborative teaching, we categorized the collected critical incidents according to the system's three terminals: the tactile terminal (hardware), the parent terminal (software), and the teacher terminal (software). Using the FBS model, we mapped observed user behaviors back to the design structure to verify the achievement of functional objectives.

The structure in Fig. 3 is presented in a matrix format and contains three main columns: Structure (Design Features), Behavior (Observed Interactions), and Function (Validated Outcomes). The rows represent the three system terminals: Student (Hardware, in teal), Parent (Software, in amber), and Teacher (Software, in indigo). Five critical incidents (Cases A-E) are mapped across these columns, demonstrating how specific structural designs (e.g., dot height, live control) trigger specific user behaviors (e.g., confident scanning, precise intervention), ultimately validating intended educational functions like independent learning and high-quality teaching intervention.

5.1 Student terminal: validation of tactile experience and independence

Validation for the student terminal focused on whether the hardware's physical properties supported efficient tactile recognition and emotional security. Table 1 presents three core cases (A, B, C), validating the effectiveness of the hardware design in terms of tactile ergonomics, interactive feedback loops, and emotional design.

During the free exploration phase, multiple younger participants (e.g., #07, #15) exhibited non-task-oriented affectionate behaviors. They not only used the device for learning but also repeatedly stroked its edges, even cradling it in their arms. One child, who had never used electronic devices before, transitioned from initial caution to active and relaxed dual exploration within the first five minutes of interaction.

This unexpected positive behavior directly validated the design value of the "rounded super-elliptical body curves" (Structure). The non-threatening physical form successfully eliminated the defensive mindset toward unfamiliar objects commonly seen in visually impaired children, triggering "active exploration and closeness" (Behavior). This fulfilled the top-ranked need in QFD weight analysis: "sense of security and emotional compensation" (Function).

5.2 Parent terminal: validation of cross-cognitive support

The verification focus for parents lies in bridging the cognitive gap where sighted parents lack Braille literacy. Case D

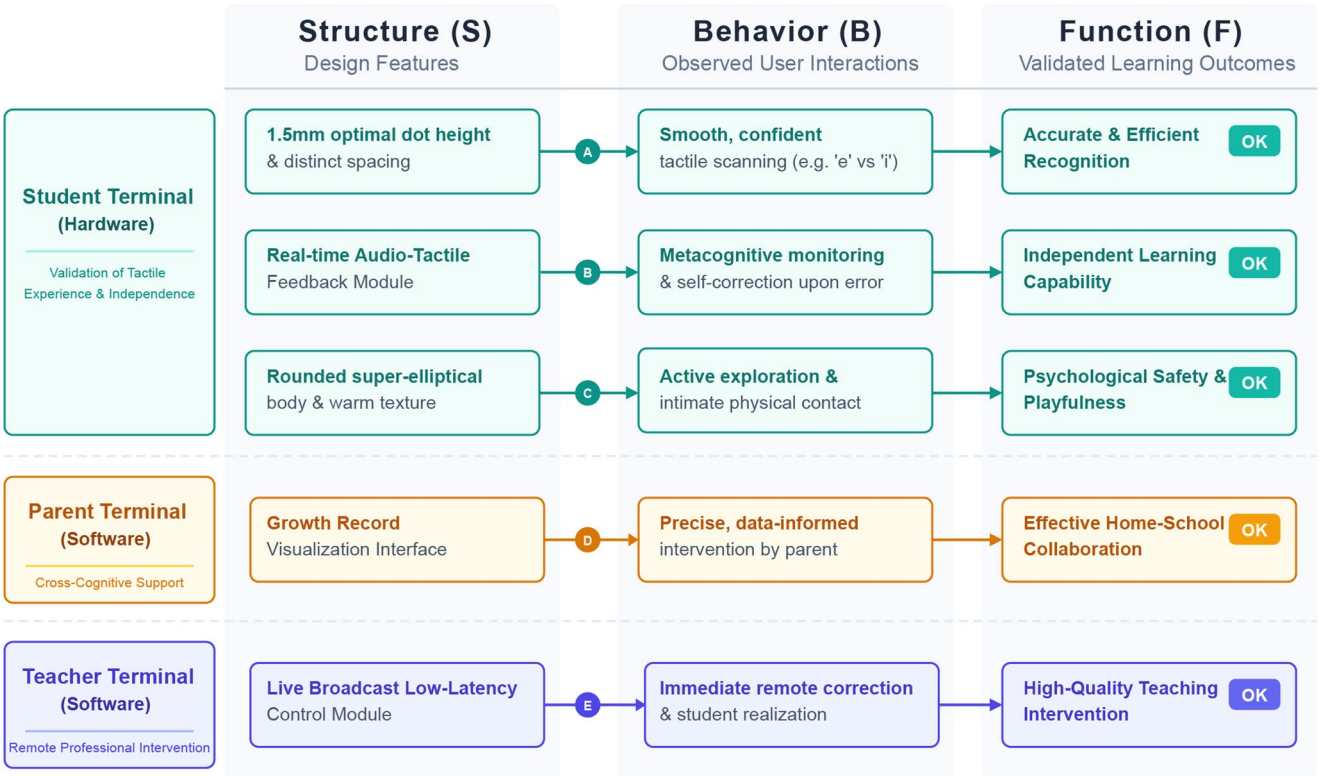


Fig. 3 The FBS mapping diagram

Table 1 Critical incident analysis for student hardware terminal

Case ID	Theme	Critical incident description (observation)	FBS mapping (structure → behavior → function)
A	**Tactile Ergonomics**	**Subject #04 (Age 12):** Distinguished between mirror-image characters 'e' and 'i' instantly (<2 s). Unlike with traditional paper Braille, the student used a single, smooth finger sweep without repetitive pressing or hesitation	**Structure:** 1.5 mm optimal dot height & distinct spacing **Behavior:** Smooth, confident tactile scanning **Function: Accurate & Efficient Recognition** (Validated)
B	**Feedback Loop**	**Subject #11 (Age 10):** Pressed 'd' instead of 'b' during dictation. The system immediately triggered an error sound. The student paused, listened to the voice prompt, and self-corrected without asking for teacher help	**Structure:** Real-time Audio-Tactile Feedback Module **Behavior:** Metacognitive monitoring & self-correction **Function: Independent Learning Capability** (Validated)
C	**Emotional Safety**	**Subject #07 (Age 6):** Initially timid, but after 5 min, began hugging the device and stroking its edges repeatedly. No "withdrawal behaviors" (common with cold, sharp mechanical devices) were observed	**Structure:** Rounded super-elliptical body & warm texture **Behavior:** Active exploration & intimate physical contact **Function: Psychological Safety & Playfulness** (Validated)

Table 2 Critical incident analysis for parent software terminal

Case ID	Theme	Critical incident description (observation)	FBS mapping (structure → behavior → function)
D	**Data-Driven Support**	**Parent #09 (No Braille Knowledge):** Checked the app and noticed a high error rate in "Pinyin b". Instead of asking vague questions, she specifically suggested: "Shall we review 'b' again?" The child then practiced that specific character	**Structure:** Growth Record Visualization Interface **Behavior:** Precise, data-informed intervention by the parent **Function: Effective Home-School Collaboration** (Validated)

Table 3 Critical incident analysis for teacher software terminal

Case ID	Theme	Critical incident description (observation)	FBS mapping (structure → behavior → function)
E	**Remote Haptic Control**	**Teacher #02:** Noticed Student #05 missed a dot via video. She clicked the control panel, and the Braille dots on the student's device refreshed instantly. The student felt the change, stopped, and said, "Ah, there is a dot here."	**Structure:** Live Broadcast Low-Latency Control Module **Behavior:** Immediate remote correction & student realization **Function: High-Quality Teaching Intervention** (Validated)

in Table 2 demonstrates how visual data empowers parental involvement in tutoring.

This incident powerfully validates the effectiveness of the "visualization of growth records" (Structure) design within the software interface. It successfully triggered parents' "data-driven precision tutoring behavior" (Behavior), breaking down the cognitive barriers between parents with normal vision and visually impaired children, thereby achieving the goal of "barrier-free home learning supervision" (Function). This directly addresses the high-priority demand from parents in the QFD matrix for "monitoring and managing the learning process."

5.3 Teacher terminal: validation of remote professional intervention

The verification focus for teachers lies in overcoming the "lack of tactile feedback" and "delayed response" inherent in remote teaching. Case E in Table 3 demonstrates how the system uses technological means to replicate the quality of in-person, hands-on instruction.

This moment marks a decisive turning point in distance education. It demonstrates that the technological breakthrough of "real-time remote control" (Structure) successfully triggered teachers' "immediate intervention behavior" (Behavior) and students' "instant corrective actions". By eliminating feedback delays caused by spatial distance, this design successfully recreates the "high-quality teaching interaction" (Function) achievable only through in-person, hands-on instruction, validating the system's core value in enhancing teaching quality.

6 Discussion

This study set out to address the "double disadvantage" faced by visually impaired children in remote areas. The lack of specialized educational resources and the inadequacy of traditional distance learning tools. By integrating

Quality Function Deployment (QFD) with the Function-Behavior-Structure (FBS) model, we developed a remote Braille learning ecosystem.

The system architecture adopts a classic bottom-up four-layer model, which comprehensively illustrates the data flow and operational mechanisms of the Braille learning system (Fig. 2). At the base, the Device Layer, centered on a single-chip microcomputer and PWM drivers, is responsible for precise physical tactile output. Data then travels through the low-latency Communication Layer (Wi-Fi/WebRTC) into the Cloud Services Layer, which centrally manages live streaming and device status. Finally, the top Applications Layer extends into two distinct terminals: the Teacher Terminal provides professional remote broadcasting and course control, while the Parental Terminal offers intuitive learning data visualization and communication feedback. The entire architecture clearly delineates a complete technological closed-loop, from foundational hardware tactile interaction to top-level, multi-terminal, cross-regional collaboration.

The discussion that follows interprets our findings through the lens of critical incidents observed during the evaluation, highlighting how specific design interventions bridge the tactile gap, empower stakeholders, and promote equity in low-resource settings.

6.1 Reinterpreting the haptic gap in remote education

Traditional distance education for the blind often fails because it relies heavily on audio, neglecting the critical role of haptic feedback in Braille literacy. Our CIT analysis reveals that bridging this "haptic gap" requires more than just displaying Braille; it demands a real-time, tactile-auditory feedback loop.

The critical incident involving Subject #11 (Case B) demonstrates this mechanism in action. When the student made an error during dictation, the system's immediate error sound triggered a self-correction behavior without teacher intervention. This validates that the "Real-time Audio-Tactile Feedback Module" (Structure) successfully substitutes the physical presence of a teacher, enabling independent error recovery (Function). Furthermore, Case Ashowed that specific physical parameters—specifically the 1.5mm dot height determined via QFD—are not merely ergonomic choices but cognitive necessities. The smooth, confident scanning behavior observed in Subject #04 confirms that precise structural affordances are prerequisite for reducing the cognitive load of "decoding" tactile information, thereby allowing students to focus on "comprehending" the content.

These findings suggest that in remote inclusive education, technology must act as a "haptic surrogate", providing the physical guidance and confirmation that a human teacher would typically offer in a classroom setting.

6.2 Empowering the parents as co-educators

A significant barrier in home-based Braille education is the "literacy gap" between blind children and their sighted parents, who often cannot read Braille. This study challenges the traditional exclusion of parents from the technical aspects of Braille instruction.

Our findings from Case D (Parent #09) illustrate a paradigm shift from "passive observers" to "active co-educators". The visualization of the growth record allowed a parent with no knowledge of Braille to identify specific learning hurdles (e.g., error rates in "Pinyin b") and initiate targeted practice. This confirms that well-designed data visualization can bridge the cognitive barrier between sighted and non-sighted stakeholders. By translating abstract Braille performance into accessible visual metrics, the system effectively democratizes the teaching process, fulfilling the social need for family engagement identified in our initial needs analysis. This "empowerment via visualization" is crucial for creating a sustainable support network in rural families where professional resources are scarce.

6.3 Design for equity in low-resource settings

The digital divide in special education is often exacerbated by the high cost of existing assistive technologies (e.g., refreshable Braille displays costing thousands of dollars). This study demonstrates that affordability does not require compromising educational quality.

The CIT analysis of Case E (Remote Haptic Control) proves that low-cost hardware, when coupled with optimized software, can achieve high-fidelity teaching interactions. The ability for a teacher to remotely refresh a student's Braille display with low latency enabled immediate pedagogical intervention, replicating the quality of face-to-face instruction. This scalability is vital for Underserved Villages. Unlike expensive solutions restricted to elite institutions, our system' s modular design—separating the costly logic (smartphone) from the tactile interface (learning machine)—offers a financially viable path for widespread adoption in developing regions. By reducing the economic threshold for access, the design moves beyond a mere product innovation to become a systemic intervention for educational equity, ensuring that geographic remoteness no longer dictates a child's educational destiny.

6.4 Limitations and future directions

While the system demonstrates significant potential, several limitations must be acknowledged:

1. Target Demographics: The current curriculum and tactile interface are optimized for preschool and early primary-aged children (foundational Braille acquisition). As noted in the limitations of our prototype, the simplified modules are unsuitable for older learners requiring advanced notation (e.g., mathematics or music).
2. Dependency Risks: The system's reliance on external devices (smartphones/computers) and stable power sources poses challenges in transient or extremely impoverished environments where such infrastructure may be inconsistent.
3. Content Scalability: Although offline learning via IC cards is supported, expanding the content library requires physical updates (new cards), which is logistically burdensome compared to cloud-based updates.
4. Sample Size: The qualitative validation relied on a small sample of 20 students. While CIT provided deep insights, future studies should employ larger-scale quantitative testing to measure long-term literacy outcomes.

Future research should focus on developing a standalone version of the device to reduce dependency on external hardware and exploring "hybrid" content delivery methods that combine the reliability of physical cards with the flexibility of cloud updates. Additionally, extending the curriculum to cover advanced subjects will be crucial for supporting visually impaired students throughout their entire academic journey.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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